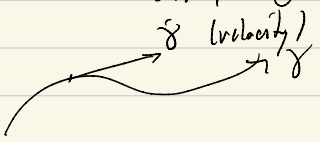
Parametrized curves in $\mathbb{R}^3$

A <u>regular curve</u> in $\mathbb{R}^3$ is a C^∞ map $\gamma : I \subset \mathbb{R} \to \mathbb{R}^3$

such that $\dot{\gamma}(t) \neq 0$ for all $t \in I$



$\left(\dot{\gamma} = \frac{d}{dt}\gamma \right)$

<u>arclength</u> $s(t) := \int_{t_0}^{t} \|\dot{\gamma}(t')\| \, dt'$ for some t_0 fixed

$\frac{ds}{dt} = \underset{\text{(speed)}}{\|\dot{\gamma}\|} > 0$ for all t

Inverse Fn. Thm $\Longrightarrow$ $s(t)$ admits a C^∞ inverse

$\Rightarrow$ <u>Lemma</u> A regular curve can be reparametrized by arclength.
("unit speed")

Example: circle of radius r, angle velocity $\omega > 0$

$\gamma(t) = (r\cos\omega t, r\sin\omega t)$, $\|\dot{\gamma}\| = r\omega$

$\Rightarrow$ arclength $s = r\omega t$, $t = s/r\omega$

$\gamma(s) = (r\cos(s/r), r\sin(s/r))$

This gives $\|\dot{\gamma}\| = 1$, $\|\ddot{\gamma}\| = 1/r$

<u>Note</u>: For a unit speed curve, differentiating $\dot{\gamma}\cdot\dot{\gamma} = 1$ gives $\dot{\gamma} \perp \ddot{\gamma}$

define: the <u>curvature</u> of a regular curve with $\|\dot{\gamma}\| = 1$

$\kappa := \|\ddot{\gamma}\|$

In general, arc length is difficult to compute explicitly

We can use the chain rule to avoid this

For $\gamma(t)$, set $v := \|\dot\gamma\| = \frac{ds}{dt}$

$$\frac{d}{dt} = \frac{ds}{dt}\frac{d}{ds} \quad\Longrightarrow\quad \frac{d}{ds} = \frac{1}{v}\frac{d}{dt}$$

$$\frac{d\gamma}{ds} = \frac{1}{v}\dot\gamma \qquad \text{(unit vector)}$$

$$\frac{d^2\gamma}{ds^2} = \frac{1}{v}\left(\frac{1}{v}\ddot\gamma - \frac{\dot v}{v^2}\dot\gamma\right)$$

$$\kappa = \frac{1}{v^3}\|v\ddot\gamma - \dot v\dot\gamma\|$$

In $\mathbb{R}^3$ we can use the cross product

$$\kappa = \frac{\|\dot\gamma\times\ddot\gamma\|}{v^3}$$

Surfaces in $\mathbb{R}^3$

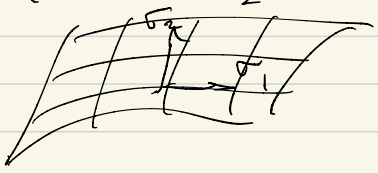
A <u>coordinate patch</u> is a C^∞ map $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ is open and connected

We'll use general coord's (u^1, u^2) for U

derivatives: $\sigma_1 := \frac{\partial\sigma}{\partial u^1}$, $\sigma_2 := \frac{\partial\sigma}{\partial u^2}$

(Einstein convention: upper/lower indices signify behavior under change of coord's)

σ is <u>regular</u> if σ_1 and σ_2 are independent at each pt.



<u>example</u>: paraboloid $z = x^2 + y^2$

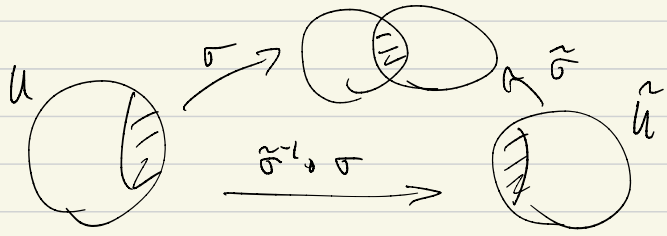
$\sigma(u,v) = (u, v, u^2+v^2) \longrightarrow \sigma_u = (1, 0, 2u)$, $\sigma_v = (0, 1, 2v)$
regular at all pts.

$\sigma(r,\theta) = (r\cos\theta, r\sin\theta, r^2)$ $\quad \sigma_r = (\cos\theta, \sin\theta, 2r)$
$\sigma_\theta = (-r\sin\theta, r\cos\theta, 0)$
$\sigma_\theta = 0$ at $r = 0 \Longrightarrow$ regular only for $r > 0$

A <u>smooth surface</u> in $\mathbb{R}^3$ is a set S covered by an <u>atlas</u> = family of regular patches
(regular $\Longrightarrow$ S has a well-defined tangent plane at each pt.)

The inverse function thm implies the following:

<u>Prop</u>: For overlapping regular patches, transition maps are diffeomorphisms



Smooth maps are defined in local coord.'s
- $f: S \to \mathbb{R}$ is smooth if $f \circ \sigma$ is C^∞ for each patch σ
- $f: S_1 \to S_2$ is smooth if $\tilde{\sigma}^{-1} \circ f \circ \sigma$ is C^∞ for pairs of patches